

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12402337
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Nishi; Koichi et al.

Semiconductor device and method for controlling semiconductor device

Abstract

When a positive gate voltage is applied to a first one of a first gate electrode and a second gate electrode, and current flows from a collector electrode to an emitter electrode, a semiconductor device applies a positive gate voltage to a second one of the first gate electrode and the second gate electrode. When a positive gate voltage is applied to the first one and current flows from the emitter electrode to the collector electrode, the semiconductor device applies voltage equal to or less than reference voltage to the second one.

Inventors: Nishi; Koichi (Tokyo, JP), Tsukuda; Masanori (Tokyo, JP), Soneda; Shinya (Tokyo, JP), Furukawa; Akihiko (Tokyo, JP)

Applicant: Mitsubishi Electric Corporation (Tokyo, JP)

Family ID: 1000008782385

Assignee: Mitsubishi Electric Corporation (Tokyo, JP)

Appl. No.: 17/845833

Filed: June 21, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230073864 A1	Mar. 09, 2023

Foreign Application Priority Data

JP	2021-144695	Sep. 06, 2021
----	-------------	---------------

Publication Classification

Int. Cl.: H10D12/00 (20250101); H10D30/66 (20250101); H10D62/17 (20250101); H10D64/27 (20250101)

U.S. Cl.:

CPC H10D12/481 (20250101); H10D30/668 (20250101); H10D62/393 (20250101); H10D64/513 (20250101);

Field of Classification Search

CPC: H10D (12/481); H10D (64/513); H10D (30/668); H10D (30/0223); H10D (30/0275); H10D (30/283); H10D (62/393); H10D (62/40); H10D (62/605)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2017/0250685	12/2016	Bina et al.	N/A	N/A
2020/0126973	12/2019	Kamibaba	N/A	H10D 84/403
2021/0091072	12/2020	Matsudai	N/A	H01L 29/8613
2021/0376167	12/2020	Soneda	N/A	H01L 29/8613
2022/0165727	12/2021	Kuroda	N/A	H10D 12/481
2022/0376605	12/2021	Rahimo	N/A	H02M 1/088

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013-251395	12/2012	JP	N/A
5768395	12/2014	JP	N/A
2017-163136	12/2016	JP	N/A

Primary Examiner: Campbell; Shaun M

Attorney, Agent or Firm: Studebaker Brackett PLLC

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present disclosure relates to a semiconductor device and a method for controlling the semiconductor device.

Description of the Background Art

(2) In the fields of general-purpose inverters, AC servomotors, and the like, an insulated gate bipolar transistor (IGBT) and a diode are used for power modules and the like that perform variable speed control of a three-phase motor from the viewpoint of energy saving. In order to reduce inverter loss, an IGBT and a diode having low switching loss and low on-voltage are required.

(3) A reverse conducting IGBT (RC-IGBT) has been proposed as a device in which an IGBT and a diode are integrated in one chip. Further, for example, Japanese Patent No. 5768395 proposes a technique of inputting separate control signals to an upper electrode and a lower electrode of a split

gate structure.

(4) However, in the related art, since a voltage is applied to a first gate electrode and a second gate electrode corresponding to the upper electrode and the lower electrode based on a recovery timing and a carrier lifetime of a diode, there has been a problem that control of voltage application is complicated.

SUMMARY

(5) The present disclosure has been made in view of the above problem, and an object of the present disclosure is to provide a technique capable of facilitating control of voltage application in a semiconductor device.

(6) A semiconductor device according to the present disclosure includes a semiconductor substrate including a first main surface and a second main surface opposite to the first main surface. The semiconductor substrate includes a drift layer of a first conductivity type provided between the first main surface and the second main surface, a carrier accumulation layer of the first conductivity type provided on the first main surface side of the drift layer, a base layer of a second conductivity type provided on the first main surface side of the carrier accumulation layer, an emitter layer of the first conductivity type selectively provided on the first main surface side of the base layer, a buffer layer of the first conductivity type provided on the second main surface side of the drift layer, and a collector layer of the second conductivity type and a cathode layer of the first conductivity type provided on the second main surface side of the buffer layer. The semiconductor device further includes a gate insulating film provided on an inner wall of a trench that penetrates the emitter layer, the base layer, and the carrier accumulation layer from the first main surface side and reaches the drift layer, a first gate electrode provided on the inner wall on the first main surface side with the gate insulating film interposed between them, and having an end portion closer to the second main surface than an end portion of the base layer on the first main surface side, a second gate electrode insulated from the first gate electrode, provided on the inner wall on the second main surface side with the gate insulating film interposed between them, and having an end portion closer to the first main surface than an end portion of the base layer on the second main surface side, an emitter electrode provided on the first main surface, a collector electrode provided on the second main surface, and a control unit, wherein in a case where the control unit applies a positive gate voltage to a first one of the first gate electrode and the second gate electrode, and current flows from the collector electrode to the emitter electrode, the control unit applies a positive gate voltage to a second one of the first gate electrode and the second gate electrode, and in a case where the control unit applies a positive gate voltage to the first one, and current flows from the emitter electrode to the collector electrode, the control unit applies a voltage equal to or less than a reference voltage to the second one.

(7) It is possible to facilitate control of voltage application in the semiconductor device.

(8) These and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a cross-sectional view illustrating a configuration of a semiconductor device according to a first preferred embodiment;

(2) FIG. 2 is a circuit diagram illustrating a connection example of a semiconductor element according to the first preferred embodiment;

(3) FIG. 3 is a timing chart illustrating operation of the semiconductor device according to the first preferred embodiment;

- (4) FIGS. **4** and **5** are each a timing chart illustrating operation of a semiconductor device according to a second preferred embodiment;
- (5) FIGS. **6** and **7** are each a cross-sectional view illustrating a configuration of a semiconductor device according to a third preferred embodiment;
- (6) FIGS. **8** and **9** are each a plan view illustrating a configuration of a semiconductor device according to a fourth preferred embodiment;
- (7) FIGS. **10** and **11** are each a plan view illustrating a configuration of a semiconductor device according to a fifth preferred embodiment;
- (8) FIGS. **12** and **13** are each a plan view illustrating a configuration of a semiconductor device according to a sixth preferred embodiment;
- (9) FIGS. **14** and **15** are each a plan view illustrating a configuration of a semiconductor device according to a seventh preferred embodiment;
- (10) FIG. **16** is a cross-sectional view illustrating a configuration of a semiconductor device according to an eighth preferred embodiment;
- (11) FIGS. **17** and **18** are each a cross-sectional view illustrating a configuration of a semiconductor device according to a ninth preferred embodiment;
- (12) FIG. **19** is a cross-sectional view illustrating a configuration of a semiconductor device according to a tenth preferred embodiment;
- (13) FIG. **20** is a cross-sectional view illustrating a configuration of a semiconductor device according to an eleventh preferred embodiment;
- (14) FIG. **21** is a cross-sectional view illustrating a configuration of a semiconductor device according to a twelfth preferred embodiment;
- (15) FIG. **22** is a cross-sectional view illustrating a configuration of a semiconductor device according to a thirteenth preferred embodiment;
- (16) FIG. **23** is a cross-sectional view illustrating a configuration of a semiconductor device according to a fourteenth preferred embodiment; and
- (17) FIG. **24** is a cross-sectional view illustrating a configuration of a semiconductor device according to a fifteenth preferred embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) Hereinafter, a preferred embodiment will be described with reference to the attached drawings. Features described in the following preferred embodiments are examples, and all features are not necessarily essential. Further, in description below, similar constituent elements in a plurality of preferred embodiments are denoted by the same or similar reference numerals, and a different constituent element will be mainly described. Further, in description below, specific positions and directions such as “upper”, “lower”, “left”, “right”, “front”, or “back” may not necessarily coincide with actual positions and directions in practice. Further, that a certain portion has a higher concentration than another portion means that, for example, an average of concentrations of the certain portion is higher than an average of concentrations of the another portion. Conversely, that a certain portion has a lower concentration than another portion means that, for example, an average of concentrations of the certain portion is lower than an average of concentrations of the another portion. Further, in description below, a first conductivity type is n-type and a second conductivity type is p-type, but the first conductivity type may be p-type and the second conductivity type may be n-type.

First Preferred Embodiment

(19) FIG. **1** is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a first preferred embodiment. Hereinafter, a case where the semiconductor element is an RC-IGBT will be described as an example.

(20) The semiconductor element of FIG. **1** includes a semiconductor substrate **50**. The semiconductor substrate **50** has a first main surface **50a** and a second main surface **50b** on the opposite side to the first main surface **50a**. The semiconductor substrate **50** includes a drift layer **1**

of a first conductivity type, a carrier accumulation layer **2** of the first conductivity type, a base layer **15** of a second conductivity type, an emitter layer **13** of the first conductivity type, a contact layer **14** of the second conductivity type, a buffer layer **3** of the first conductivity type, a collector layer **16** of the second conductivity type, and a cathode layer **26** of the first conductivity type.

(21) The drift layer **1** is provided between the first main surface **50a** and the second main surface **50b**. The carrier accumulation layer **2** is provided on the first main surface **50a** side of the drift layer **1**. For example, an impurity concentration of the first conductivity type of the carrier accumulation layer **2** is higher than an impurity concentration of the first conductivity type of the drift layer **1**.

(22) The base layer **15** is provided on the first main surface **50a** side of the carrier accumulation layer **2**. The emitter layer **13** is selectively provided on the first main surface **50a** side of the base layer **15**. The contact layer **14** is selectively provided on the first main surface **50a** side of the base layer **15** and is adjacent to the emitter layer **13**. For example, an impurity concentration of the second conductivity type of the contact layer **14** is higher than an impurity concentration of the second conductivity type of the base layer **15**. Note that a portion of the contact layer **14** in FIG. **1** may be the base layer **15**.

(23) The buffer layer **3** is provided on the second main surface **50b** side of the drift layer **1**. For example, an impurity concentration of the first conductivity type of the buffer layer **3** is higher than an impurity concentration of the first conductivity type of the drift layer **1**. The collector layer **16** is selectively provided on the second main surface **50b** side of the buffer layer **3**. The cathode layer **26** is selectively provided on the second main surface **50b** side of the buffer layer **3** and is adjacent to the collector layer **16**. For example, an impurity concentration of the first conductivity type of the cathode layer **26** is higher than an impurity concentration of the first conductivity type of the buffer layer **3**.

(24) The semiconductor element in FIG. **1** includes a gate insulating film **11a**, a first gate electrode **11b**, a second gate electrode **11c**, an interlayer insulating film **4**, an emitter electrode **6**, and a collector electrode **7**. The gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c** constitute an active trench **11**.

(25) The gate insulating film **11a** is provided on an inner wall of a trench that penetrates the emitter layer **13**, the base layer **15**, and the carrier accumulation layer **2** from the first main surface **50a** side and reaches the drift layer **1**.

(26) The first gate electrode **11b** is provided on an inner wall of the trench on the first main surface **50a** side with the gate insulating film **11a** interposed between them. The first gate electrode **11b** has an end portion closer to the second main surface **50b** than an end portion of the base layer **15** on the first main surface **50a** side.

(27) The second gate electrode **11c** is provided on an inner wall of the trench on the second main surface **50b** side with the gate insulating film **11a** interposed between them. The second gate electrode **11c** is insulated from the first gate electrode **11b** by, for example, the gate insulating film **11a**. The second gate electrode **11c** has an end portion closer to the first main surface **50a** than an end portion of the base layer **15** on the second main surface **50b** side.

(28) The interlayer insulating film **4** is provided on the first main surface **50a** of the semiconductor substrate **50**, and is provided with an opening portion for exposing the contact layer **14**. The emitter electrode **6** is provided on the first main surface **50a** of the semiconductor substrate **50** and the interlayer insulating film **4**, and is electrically connected to the contact layer **14** in the opening portion of the interlayer insulating film **4**.

(29) The collector electrode **7** is provided on the second main surface **50b** of the semiconductor substrate **50**.

(30) One or more sets of the emitter layer **13**, the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c** are provided on the first main surface **50a** side, and are provided in an IGBT region **10** corresponding to the collector layer **16** in plan view. In the IGBT

region **10**, an IGBT is realized by the drift layer **1**, the emitter electrode **6**, the collector electrode **7**, the gate insulating film **11a**, the first gate electrode **11b**, the second gate electrode **11c**, the emitter layer **13**, the base layer **15**, the collector layer **16**, and the like. In a diode region **20** corresponding to the cathode layer **26** in plan view, a diode is realized by the drift layer **1**, the collector electrode **7**, the contact layer **14**, the base layer **15**, the cathode layer **26**, and the like.

(31) FIG. **2** is a circuit diagram illustrating a connection example of the semiconductor element of FIG. **1**. FIG. **2** illustrates an example in which two RC-IGBTs each of which is the semiconductor element of FIG. **1** are fully bridge-connected to a power supply V_{cc} , and load current flows through an inductor which is a load Load.

(32) Hereinafter, in order to distinguish the two semiconductor elements in FIG. **1**, the semiconductor element on the P side is referred to as a semiconductor element RCp, and the semiconductor element on the N side is referred to as a semiconductor element RCn. Further, collector current flowing through each of the semiconductor elements RCp and RCn is positive in the direction in which the current flows from the collector electrode **7** toward the emitter electrode **6**. Collector current, a first gate electrode voltage, and a second gate electrode voltage of the semiconductor element RCp are referred to as I_{cp} , V_{gep1} , and V_{gep2} , respectively, and collector current, a first gate electrode voltage, and a second gate electrode voltage of the semiconductor element RCn are referred to as I_{cn} , V_{gen1} , and V_{gen2} , respectively.

(33) The semiconductor device according to the first preferred embodiment includes a control unit **51**. The control unit **51** is realized by, for example, a current detection device, a central processing unit (CPU), and the like. The control unit **51** applies a positive gate voltage to a first one of the first gate electrode **11b** and the second gate electrode **11c**. Further, based on the direction of the current flowing between the emitter electrode **6** and the collector electrode **7**, the control unit **51** applies a positive gate voltage or a voltage equal to or less than a reference voltage to a second one of the first gate electrode **11b** and the second gate electrode **11c**. The voltage equal to or less than the reference voltage here includes a negative gate voltage or a reference voltage (corresponding to 0 V).

(34) Hereinafter, it is assumed that the first one of the gate electrodes is the first gate electrode **11b** and the second one is the second gate electrode **11c**, but the first one may be the second gate electrode **11c** and the second one may be the first gate electrode **11b**. Note that the two control units **51** in FIG. **2** may be realized by one control unit.

(35) FIG. **3** is a timing chart illustrating application control by the control unit **51** according to the first preferred embodiment. FIG. **3** illustrates an example in which a control signal (for example, a PWM signal) is input to the first gate electrode **11b** of the semiconductor elements RCn and RCp, the semiconductor element RCn on the N side operates as an IGBT, and the semiconductor element RCp on the P side operates as a diode.

(36) The control unit **51** inputs a positive gate control signal voltage, which is a positive gate voltage, to the first gate electrode **11b** of the semiconductor elements RCp and RCn at different timings, like V_{gep1} and V_{gen1} . Note that, as V_{gen1} and V_{gep1} , a delay of dead time t_1 may be provided between the semiconductor element RCp and the semiconductor element RCn with respect to input of the positive gate control signal voltage.

(37) In the semiconductor element RCn on the N side, when the positive gate control signal voltage is input to the first gate electrode **11b** and the collector current is in the positive direction (that is, $I_{cn} > 0$), the control unit **51** inputs the positive gate control signal voltage to the second gate electrode **11c** as in V_{gen2} . By such control, an inversion layer of the first conductivity type is formed on the base layer **15** adjacent to the first gate electrode **11b** and the second gate electrode **11c**, and the emitter layer **13** is electrically connected to the carrier accumulation layer **2** and the drift layer **1**, so that the semiconductor element RCn on the N side operates as an IGBT. As in V_{gen1} and V_{gen2} , a delay of time t_2 may be provided between input of the positive gate control signal voltage to the second gate electrode **11c** and input of the positive gate control signal voltage

to the first gate electrode **11b**.

(38) In the semiconductor element RCp on the P side, when the positive gate control signal voltage is input to the first gate electrode **11b** and the collector current is in the negative direction (that is, $I_{cp} < 0$), the control unit **51** inputs a voltage equal to or less than the reference voltage to the second gate electrode **11c** as Vgep2. By such control, the inversion layer of the first conductivity type is not formed on the base layer **15** adjacent to the second gate electrode **11c**, and the emitter layer **13** is not electrically connected to the carrier accumulation layer **2** and the drift layer **1**, so that the semiconductor element RCp on the P side operates as a diode. Note that since carriers supplied from the cathode layer **26** are accumulated in the drift layer **1** due to such electrical non-conduction, an on-voltage VF of a diode can be reduced.

(39) Although not illustrated, with respect to the semiconductor element RCn on the N side when the positive gate control signal voltage is input to the first gate electrode **11b** and the collector current is in the negative direction (that is, $I_{cn} < 0$), the control unit **51** inputs a voltage equal to or less than the reference voltage to the second gate electrode **11c**. For this reason, at this time, the semiconductor element RCn on the N side operates as a diode. With respect to the semiconductor element RCp on the P side, when the positive gate control signal voltage is input to the first gate electrode **11b** and the collector current is in the positive direction (that is, $I_{cp} > 0$), the control unit **51** inputs the positive gate control signal voltage to the second gate electrode **11c**. For this reason, at this time, the semiconductor element RCp on the P side operates as an IGBT.

Summary of First Preferred Embodiment

(40) In the semiconductor device according to the first preferred embodiment, when the control unit **51** applies a positive gate voltage to the first gate electrode **11b**, and current flows from the collector electrode **7** to the emitter electrode **6**, the control unit **51** applies a positive gate voltage to the second gate electrode **11c**. In contrast, when the control unit **51** applies a positive gate voltage to the first gate electrode **11b**, and current flows from the emitter electrode **6** to the collector electrode **7**, the control unit **51** applies a voltage equal to or less than the reference voltage to the second gate electrode **11c**.

(41) According to such a configuration, the voltage input to the second gate electrode **11c** is uniquely determined only by the voltage input to the first gate electrode **11b** and whether the collector current is positive or negative. For this reason, it is not necessary to control the voltage application on the basis of a recovery timing of a diode or a carrier lifetime, and thus, it is possible to facilitate the control of the voltage application in the semiconductor device. Further, when the semiconductor element operates as a diode, carriers supplied from the cathode layer **26** are accumulated in the drift layer **1**, so that the on-voltage VF of the diode can be reduced.

Second Preferred Embodiment

(42) FIG. **4** is a timing chart illustrating application control of the control unit **51** according to a second preferred embodiment. The control unit **51** according to the second preferred embodiment performs application control similar to that of the control unit **51** according to the first preferred embodiment. However, the control unit **51** according to the second preferred embodiment applies a positive gate voltage to both the first gate electrode **11b** and the second gate electrode **11c** before the voltage applied to the first gate electrode **11b** is switched from a positive gate voltage to a voltage equal to or less than the reference voltage, regardless of whether the collector current is positive or negative.

(43) In the example of FIG. **4**, a positive gate control signal voltage is input to the first gate electrode **11b** and the second gate electrode **11c** of the semiconductor element RCp on the P side in a certain period t3 before recovery of the semiconductor element RCp on the P side, such as Vgep1 and Vgep2. As a result, during the certain period t3, an inversion layer of the first conductivity type is formed on the base layer **15** adjacent to the first gate electrode **11b** and the second gate electrode **11c** of the semiconductor element RCp on the P side, and the emitter layer **13**, the carrier accumulation layer **2**, and the drift layer **1** electrically conduct.

(44) According to the semiconductor device according to the second preferred embodiment that performs such control, carriers in the drift layer **1** can be discharged to the emitter electrode **6**, so that recovery loss can be reduced. Since the recovery loss and the on-voltage V_F are in a trade-off relationship, the on-voltage V_F of the semiconductor element RCp on the P side increases in the certain period t_3 . According to the second preferred embodiment, the trade-off relationship can be adjusted by adjusting the certain period t_3 .

(45) FIG. 5 is a timing chart illustrating another application control of the control unit **51** according to the second preferred embodiment. In the example of FIG. 5, a positive gate control signal voltage is applied to the second gate electrode **11c** from the certain period t_3 to a certain period t_4 after a voltage equal to or less than the reference voltage is applied to the first gate electrode **11b**. By such control, recovery loss can be reduced as compared with that in the application control of FIG. 4. Here, during the certain period t_4 , an inversion layer of the first conductivity type is not formed on the base layer **15** adjacent to the first gate electrode **11b** of the semiconductor element RCp on the P side, and the emitter layer **13**, the carrier accumulation layer **2**, and the drift layer **1** do not electrically conduct. For this reason, it is possible to suppress occurrence of a short circuit between the semiconductor element RCp on the P side and the semiconductor element RCn on the N side.

Third Preferred Embodiment

(46) FIG. 6 is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a third preferred embodiment. In the third preferred embodiment, a plurality of sets of the emitter layer **13**, the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c** are provided on the first main surface **50a** side of the semiconductor substrate **50**. That is, a plurality of IGBT cells A is provided on the first main surface **50a** side of the semiconductor substrate **50**. According to such a configuration, since it is possible to increase a region that operates with an IGBT, it is possible to increase channel density and increase saturation current. The plurality of IGBT cells A may be provided on the entire first main surface **50a** of the semiconductor substrate **50**.

(47) In the third preferred embodiment, a width W_1 of the IGBT region **10** is larger than 2.1 times a distance T_1 between the trench of the active trench **11** and the second main surface **50b**.

(48) When the semiconductor element operates as an IGBT by input of a positive gate control signal voltage to the first gate electrode **11b** and the second gate electrode **11c**, carriers injected from the emitter layer **13** flow into the drift layer **1** through an inversion layer adjacent to the active trench **11**. A part of the carriers that flow in is accumulated in the drift layer **1**, but a part of the other carriers is discharged from the cathode layer **26**. When the number of carriers discharged from the cathode layer **26** is large, the resistance of the drift layer **1** increases, and snapback may occur. Therefore, the width W_1 of the IGBT region **10** needs to be equal to or more than a certain value so that carriers are sufficiently accumulated in the drift layer **1**.

(49) FIG. 7 is a cross-sectional view for explaining the width W_1 that is appropriate. Here, it is assumed that the width of the active trench **11** is ignored, and it is assumed that carriers spread 45° from a bottom portion of one active trench **11** toward the collector electrode **7**. Assuming as above, a width by which the carrier spreads on the second main surface **50b** side is $2 \times T_1$. When the width W_1 is 2.1 times or more the distance T_1 , substantially all the carriers supplied from at least one active trench **11** can be accumulated in the drift layer **1**. According to the third preferred embodiment, since the width W_1 is larger than 2.1 times the distance T_1 , it is possible to suppress snapback when the semiconductor element operates as an IGBT.

Fourth Preferred Embodiment

(50) FIGS. 8 and 9 are plan views illustrating a configuration of a semiconductor element included in a semiconductor device according to a fourth preferred embodiment. In the fourth preferred embodiment, in plan view, the IGBT region **10**, the diode region **20**, a pad region **40** excluding these regions, and a termination region **30** surrounding the IGBT region **10**, the diode region **20**,

and the pad region **40** are provided on the semiconductor substrate **50**. Further, a pad **41** is provided in the pad region **40**.

(51) The diode region **20** has a quadrangular shape in plan view. The diode regions **20** having a quadrangular shape may be provided in a stripe shape as illustrated in FIG. **8**, or may be provided in an island shape as illustrated in FIG. **9**. According to the present fourth preferred embodiment, when a chip outer shape is a quadrangle, the diode region **20** can be provided in accordance with the chip outer shape. In this manner, since the width **W1** of the IGBT region **10** can be made uniform, current variation in the chip when the semiconductor element operates as an IGBT can be reduced.

Fifth Preferred Embodiment

(52) FIGS. **10** and **11** are plan views illustrating a configuration of a semiconductor element included in a semiconductor device according to a fifth preferred embodiment. In the fifth preferred embodiment, as in the fourth preferred embodiment, the IGBT region **10**, the diode region **20**, the termination region **30**, and the pad region **40** are provided on the semiconductor substrate **50** in plan view. In the present fifth preferred embodiment, the diode region **20** has a circular shape in plan view. According to the present fifth preferred embodiment as described above, current concentration at an end portion of the diode region **20** can be suppressed. The same applies to a configuration in which the diode region **20** has a polygonal shape of a pentagon or more in plan view.

Sixth Preferred Embodiment

(53) FIGS. **12** and **13** are plan views illustrating a configuration of a semiconductor element included in a semiconductor device according to a sixth preferred embodiment. In the present sixth preferred embodiment, at least a part of the IGBT region **10** is surrounded by the diode region **20** in plan view. At least a part of the IGBT region **10** may be surrounded by the single diode region **20** as illustrated in FIG. **12**, or may be surrounded by the double diode region **20** as illustrated in FIG. **13**.

(54) Here, heat generated when the semiconductor element operates as an IGBT can be released from a boundary between the IGBT region **10** and the diode region **20** to the diode region **20**. In contrast, heat generated when the semiconductor element operates as a diode can be released from the boundary to the IGBT region **10**. According to the sixth preferred embodiment, since a length of a boundary line between the IGBT region **10** and the diode region **20** can be made large, the heat dissipation property as described above can be improved.

Seventh Preferred Embodiment

(55) FIGS. **14** and **15** are plan views illustrating a configuration of a semiconductor element included in a semiconductor device according to a seventh preferred embodiment. In the present seventh preferred embodiment, at least a part of the IGBT region **10** has, for example, a shape radially expanding from the center of the chip toward an outer periphery. As illustrated in FIG. **14**, in plan view, the IGBT region **10** may include a quadrangular portion provided at the center of the chip and a portion having a shape radially expanding from the quadrangular portion toward the outer periphery. As illustrated in FIG. **15**, the IGBT region **10** in plan view may include a portion having an edge of a quadrangular shape along the outer periphery of the chip and a portion having a shape radially expanding from the center of the chip toward the outer periphery.

(56) According to the seventh preferred embodiment, similarly to the sixth preferred embodiment, the length of the boundary line between the IGBT region **10** and the diode region **20** can be made large, so that the heat dissipation property can be improved.

Eighth Preferred Embodiment

(57) FIG. **16** is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to an eighth preferred embodiment. In the eighth preferred embodiment, a plurality of sets of the emitter layer **13**, the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c** are provided on the first main surface

50a side of the semiconductor substrate **50**, and the plurality of IGBT cells **A** are provided in the IGBT region **10** in plan view.

(58) Further, in the eighth preferred embodiment, the IGBT region **10** includes a main IGBT region **10a** that is a first IGBT region and a sense IGBT region **10b** that is a second IGBT region. The sense IGBT region **10b** is adjacent to the main IGBT region **10a** and has an area smaller than that of the main IGBT region **10a**. The sense IGBT region **10b** is provided with a configuration similar to a configuration provided in the main IGBT region **10a**, and is provided with an IGBT for detecting current flowing through the IGBT in the main IGBT region **10a**.

(59) In the eighth preferred embodiment, the diode region **20** includes a main diode region **20a** that is a first diode region and a sense diode region **20b** that is a second diode region. The sense diode region **20b** has an area smaller than that of the main diode region **20a**. The sense diode region **20b** is provided with a configuration similar to a configuration provided in the main diode region **20a**, and is provided with a diode for detecting current flowing through a diode of the main diode region **20a**.

(60) The emitter electrode **6** includes a main emitter electrode **6a** that is a first emitter electrode and a sense emitter electrode **6b** that is a second emitter electrode. The main emitter electrode **6a** is provided in the main IGBT region **10a** and the main diode region **20a**. The sense emitter electrode **6b** is provided in the sense IGBT region **10b** and the sense diode region **20b**, and is separated from the main emitter electrode **6a**.

(61) According to the present eighth preferred embodiment, the sense IGBT region **10b** can detect collector current in a positive direction flowing through the main IGBT region **10a**, and the sense diode region **20b** can detect collector current in a negative direction flowing through the main diode region **20a**. Further, since the sense IGBT region **10b** and the main IGBT region **10a** are adjacent to each other, a width of the effective collector layer **16** on the second main surface **50b** side of the sense IGBT region **10b** can be increased, and as a result, snapback can be suppressed.

Ninth Preferred Embodiment

(62) FIGS. **17** and **18** are cross-sectional views illustrating a configuration of a semiconductor element included in a semiconductor device according to a ninth preferred embodiment. The semiconductor substrate **50** of the semiconductor element according to the present ninth preferred embodiment is provided with the pad region **40** excluding the IGBT region **10** and the diode region **20** in plan view as in the fourth preferred embodiment. In the pad region **40**, a first gate pad **41a** which is a gate pad illustrated in FIG. **17** is provided, and a second gate pad **41b** which is a gate pad illustrated in FIG. **18** is provided.

(63) As illustrated in FIG. **17**, the first gate electrode **11b** and the first gate pad **41a** are electrically connected by a first gate wiring **46a** and a first built-in gate resistor **45a** that is a built-in gate resistor covered with the interlayer insulating film **4**. Similarly, as illustrated in FIG. **18**, the second gate electrode **11c** and the second gate pad **41b** are electrically connected by a second gate wiring **46b** and a second built-in gate resistor **45b** that is a built-in gate resistor covered with the interlayer insulating film **4**.

(64) According to the present ninth preferred embodiment, it is possible to reduce an external gate resistor. In FIGS. **17** and **18**, the first built-in gate resistor **45a** and the second built-in gate resistor **45b** are provided on the semiconductor substrate **50**, but may be embedded inside the semiconductor substrate **50**. Further, it is not necessary to provide both the configuration of FIG. **17** and the configuration of FIG. **18**, and one of the configuration of FIG. **17** and the configuration of FIG. **18** may not be provided.

Tenth Preferred Embodiment

(65) FIG. **19** is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a tenth preferred embodiment. In the present tenth preferred embodiment, a length ($=L1$) of the first gate electrode **11b** is shorter than a length ($=L5-L2$) of the second gate electrode **11c** in a thickness direction of the semiconductor substrate

50.

(66) In the example of FIG. 19, L1 is shorter than a length (=L2) from the first main surface 50a to an end portion of the second gate electrode 11c on the first main surface 50a side. L2 is shorter than a length (=L3) from the first main surface 50a to the end portion of the base layer 15 on the second main surface 50b side. L3 is shorter than a length (=L4) from the first main surface 50a to an end portion of the carrier accumulation layer 2 on the second main surface 50b side. L4 is shorter than a length (=L5) from the first main surface 50a to an end portion of the second gate electrode 11c on the second main surface 50b side. By setting $L1 < L2 < L3$, the operation described in the first preferred embodiment becomes possible. Further, by setting $L4 < L5$, withstand voltage can be improved.

(67) Note that the shorter the L1 and the longer the L5, the thicker the carrier accumulation layer 2 can be. Therefore, by setting $L1 < L5 - L2$, the carrier accumulation layer 2 can be made thick, so that an on-voltage VCEsat of an IGBT can be reduced.

Eleventh Preferred Embodiment

(68) FIG. 20 is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to an eleventh preferred embodiment. In the present eleventh preferred embodiment, an end portion of the first gate electrode 11b on the second main surface 50b side is closer to the second main surface 50b than the end portion of the second gate electrode 11c on the first main surface 50a side. That is, $L1 > L2$ holds between L1 and L2 in FIG. 20. Here, the end portion of the first gate electrode 11b on the second main surface 50b side and the end portion of the second gate electrode 11c on the first main surface 50a side are closer to the second main surface 50b than the end portion of the base layer 15 on the first main surface 50a side, and are closer to the first main surface 50a than the end portion of the base layer 15 on the second main surface 50b side.

(69) According to the present eleventh preferred embodiment as described above, the first gate electrode 11b and the second gate electrode 11c overlap in the thickness direction of the semiconductor substrate 50. For this reason, the inversion layer formed on the base layer 15 adjacent to first gate electrode 11b and second gate electrode 11c can be stabilized.

Twelfth Preferred Embodiment

(70) FIG. 21 is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a twelfth preferred embodiment. The semiconductor element of FIG. 21 includes an insulating film 12a, a first electrode 12b, and a second electrode 12c corresponding to the gate insulating film 11a, the first gate electrode 11b, and the second gate electrode 11c, respectively, in addition to the constituent element of the semiconductor element of FIG. 1. Except that the first electrode 12b and the second electrode 12c are electrically connected to the emitter electrode 6, the insulating film 12a, the first electrode 12b, and the second electrode 12c are substantially the same as the gate insulating film 11a, the first gate electrode 11b, and the second gate electrode 11c, respectively. Due to the difference in connection, while the gate insulating film 11a, the first gate electrode 11b, and the second gate electrode 11c constitute the active trench 11, the insulating film 12a, the first electrode 12b, and the second electrode 12c constitute a dummy trench 12.

(71) The semiconductor substrate 50 in FIG. 21 includes a dummy mesa 18 corresponding to an active mesa 17 including the carrier accumulation layer 2, the base layer 15, the emitter layer 13, and the contact layer 14. Note that the active mesa 17 is included in the concept of a first stacked structure, and the dummy mesa 18 is included in the concept of a second stacked structure.

(72) The active mesa 17 is adjacent to the active trench 11 including the gate insulating film 11a, whereas the dummy mesa 18 is adjacent to the dummy trench 12 including the insulating film 12a. Except for this point, the dummy mesa 18 is substantially the same as the active mesa 17. As illustrated in FIG. 21, the dummy mesa 18 may be connected to the emitter electrode 6 via the opening portion provided in the interlayer insulating film 4. Although not illustrated, the

configuration may be such that the opening portion is not provided in the interlayer insulating film **4**, and the potential of the dummy mesa **18** is floating potential.

(73) According to the present twelfth preferred embodiment as described above, the capacitance can be reduced by the dummy trench **12**. Further, in the present twelfth preferred embodiment, in the active mesa **17** and the dummy mesa **18**, diffusion layers such as the carrier accumulation layer **2**, the base layer **15**, the emitter layer **13**, and the contact layer **14** are substantially the same. For this reason, the active trench **11** and the dummy trench **12** can be formed in a different manner depending on whether or not the connection target of the first gate electrode **11b** and the second gate electrode **11c** is changed to the emitter electrode **6**. Further, since they can be formed in a different manner only by, for example, changing a contact pattern, productivity can be improved.

Thirteenth Preferred Embodiment

(74) FIG. **22** is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a thirteenth preferred embodiment. The semiconductor element of FIG. **22** includes an insulating film **19a**, a first electrode **19b**, and a second electrode **19c** corresponding to the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c**, respectively, in addition to the constituent element of the semiconductor element of FIG. **1**. A first one of the first electrode **19b** and the second electrode **19c** is electrically connected to the emitter electrode **6**, and a second one of the first electrode **19b** and the second electrode **19c** is electrically connected to the first gate electrode **11b**. Except for this point, the insulating film **19a**, the first electrode **19b**, and the second electrode **19c** are substantially the same as the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c**, respectively. Due to the difference in connection, while the gate insulating film **11a**, the first gate electrode **11b**, and the second gate electrode **11c** constitute the active trench **11**, the insulating film **19a**, the first electrode **19b**, and the second electrode **19c** constitute a dummy active trench **19**.

(75) The semiconductor substrate **50** in FIG. **22** includes the dummy mesa **18** corresponding to the active mesa **17** including the carrier accumulation layer **2**, the base layer **15**, the emitter layer **13**, and the contact layer **14**. Note that the active mesa **17** is included in the concept of the first stacked structure, and the dummy mesa **18** is included in the concept of the second stacked structure.

(76) The active mesa **17** is adjacent to the active trench **11** including the gate insulating film **11a**, whereas the dummy mesa **18** is adjacent to the dummy active trench **19** including the insulating film **19a**. Except for this point, the dummy mesa **18** is substantially the same as the active mesa **17**.

(77) According to the present thirteenth preferred embodiment as described above, a capacitance ratio, that is, feedback capacitance/input capacitance can be adjusted by the dummy active trench **19**. For example, when the first electrode **19b** is connected to the emitter electrode **6** and the second electrode **19c** is connected to the first gate electrode **11b** of the active trench **11**, feedback capacitance is increased by the second electrode **19c**, so that the capacitance ratio can be increased. Conversely, when the first electrode **19b** is connected to the first gate electrode **11b** of the active trench **11** and the second electrode **19c** is connected to the emitter electrode **6**, input capacitance is increased by the first electrode **19b**, so that the capacitance ratio can be reduced.

Fourteenth Preferred Embodiment

(78) FIG. **23** is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a fourteenth preferred embodiment. In addition to the first preferred embodiment, the semiconductor element in FIG. **23** is provided with a contact trench **50c** that penetrates the emitter layer **13** from the first main surface **50a** side of the active mesa **17** and reaches the base layer **15**. A part of the emitter electrode **6** is provided in the contact trench **50c**, and the contact layer **14** is provided at a bottom portion of the contact trench **50c**. According to the present fourteenth preferred embodiment as described above, it is possible to enhance hole extraction performance at the time of turn-off of the semiconductor element operating as an IGBT, and thus, it is possible to improve latch-up tolerance.

Fifteenth Preferred Embodiment

(79) FIG. 24 is a cross-sectional view illustrating a configuration of a semiconductor element included in a semiconductor device according to a fifteenth preferred embodiment. In the semiconductor element of FIG. 24, the contact layer 14 is not provided at the bottom portion of the contact trench 50c in the fourteenth preferred embodiment, and a part of the emitter electrode 6 is in contact with the base layer 15 at a side wall and the bottom portion of the contact trench 50c. According to the present fifteenth preferred embodiment as described above, since the contact layer 14 having high concentration is not provided, carrier injection efficiency from the first main surface 50a side of the semiconductor element operating as a diode can be reduced, and recovery loss can be reduced.

(80) <Variation>

(81) Various developments may be performed in the preferred embodiments described above. For example, the preferred embodiments described above can be applied without limitation to a withstand voltage class of the semiconductor element and whether the semiconductor substrate 50 is any of an FZ substrate/an MCZ substrate/an epitaxial substrate. Further, a material of the semiconductor substrate 50 may be normal silicon (Si) or a wide band gap semiconductor of silicon carbide (SiC), gallium nitride (GaN), diamond or the like. In a case where the material of the semiconductor substrate 50 is a wide band gap semiconductor, stable operation under high temperature and high voltage, and high switching speed can be achieved. Further, different preferred embodiments can be combined, and a certain preferred embodiment can be partially applied to another preferred embodiment.

(82) Note that the preferred embodiments and the variations can be freely combined, and the preferred embodiments and the variations can be appropriately modified or omitted.

(83) While the disclosure has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised.

Claims

1. A semiconductor device comprising a semiconductor substrate including a first main surface and a second main surface opposite to the first main surface, wherein the semiconductor substrate includes a drift layer of a first conductivity type provided between the first main surface and the second main surface, a carrier accumulation layer of the first conductivity type provided on the first main surface side of the drift layer, a base layer of a second conductivity type provided on the first main surface side of the carrier accumulation layer, an emitter layer of the first conductivity type selectively provided on the first main surface side of the base layer, a buffer layer of the first conductivity type provided on the second main surface side of the drift layer, and a collector layer of the second conductivity type and a cathode layer of the first conductivity type provided on the second main surface side of the buffer layer, the semiconductor device further comprising: a gate insulating film provided on an inner wall of a trench that penetrates the emitter layer, the base layer, and the carrier accumulation layer from the first main surface side and reaches the drift layer; a first gate electrode provided on the inner wall on the first main surface side with the gate insulating film interposed therebetween, and having an end portion closer to the second main surface than an end portion of the base layer on the first main surface side; a second gate electrode insulated from the first gate electrode, provided on the inner wall on the second main surface side with the gate insulating film interposed therebetween, and having an end portion closer to the first main surface than an end portion of the base layer on the second main surface side; an emitter electrode provided on the first main surface; a collector electrode provided on the second main surface; and a control unit configured to apply gate voltages to the first gate electrode and the second gate electrode and configured to determine whether current flows from the collector electrode to the emitter electrode or from the emitter electrode to the collector electrode, wherein in a case where the control unit

applies a positive gate voltage to a first one of the first gate electrode and the second gate electrode, and the control unit determines that current flows from the collector electrode to the emitter electrode, the control unit is configured to apply a positive gate voltage to a second one of the first gate electrode and the second gate electrode, and in a case where the control unit applies a positive gate voltage to the first one, and the control unit determines that current flows from the emitter electrode to the collector electrode, the control unit is configured to apply a voltage equal to or less than a reference voltage to the second one.

2. The semiconductor device according to claim 1, wherein the control unit is configured to apply a positive gate voltage to both the first one and the second one before a voltage applied to the first one is switched from a positive gate voltage to a voltage equal to or less than the reference voltage.

3. The semiconductor device according to claim 1, wherein a plurality of sets of the emitter layer, the gate insulating film, the first gate electrode, and the second gate electrode are provided on the first main surface side, and a width of an IGBT region corresponding to the collector layer in plan view is larger than 2.1 times a distance between the trench and the second main surface.

4. The semiconductor device according to claim 1, wherein a diode region corresponding to the cathode layer in plan view has a quadrangular shape.

5. The semiconductor device according to claim 1, wherein a diode region corresponding to the cathode layer in plan view has a polygonal shape of a pentagon or more, or a circular shape.

6. The semiconductor device according to claim 1, wherein at least a part of an IGBT region corresponding to the collector layer is surrounded by a diode region corresponding to the cathode layer in plan view.

7. The semiconductor device according to claim 1, wherein at least a part of an IGBT region corresponding to the collector layer in plan view has a radially expanding shape.

8. The semiconductor device according to claim 1, wherein a plurality of sets of the emitter layer, the gate insulating film, the first gate electrode, and the second gate electrode are provided on the first main surface side, an IGBT region corresponding to the collector layer in plan view includes a first IGBT region and a second IGBT region that is smaller in area than the first IGBT region and is adjacent to the first IGBT region, a diode region corresponding to the cathode layer in plan view includes a first diode region and a second diode region smaller in area than the first diode region, and the emitter electrode includes a first emitter electrode provided in the first IGBT region and the first diode region, and a second emitter electrode provided in the second IGBT region and the second diode region and separated from the first emitter electrode.

9. The semiconductor device according to claim 1, further comprising: a gate pad provided in a pad region of the semiconductor substrate excluding an IGBT region corresponding to the collector layer and a diode region corresponding to the cathode layer in plan view; and a built-in gate resistor electrically connecting the gate pad and the first gate electrode or the second gate electrode.

10. The semiconductor device according to claim 1, wherein a length of the first gate electrode is shorter than a length of the second gate electrode in a thickness direction of the semiconductor substrate.

11. The semiconductor device according to claim 1, wherein the end portion of the first gate electrode on the second main surface side is closer to the second main surface than the end portion of the second gate electrode on the first main surface side.

12. The semiconductor device according to claim 1, further comprising an insulating film, a first electrode, and a second electrode respectively corresponding to the gate insulating film, the first gate electrode, and the second gate electrode, wherein the semiconductor substrate further includes a second stacked structure corresponding to a first stacked structure including the carrier accumulation layer, the base layer, and the emitter layer and adjacent to the insulating film, and the first electrode and the second electrode are electrically connected to the emitter electrode.

13. The semiconductor device according to claim 1, further comprising an insulating film, a first electrode, and a second electrode respectively corresponding to the gate insulating film, the first

gate electrode, and the second gate electrode, wherein the semiconductor substrate further includes a second stacked structure corresponding to a first stacked structure including the carrier accumulation layer, the base layer, and the emitter layer and adjacent to the insulating film, a first one of the first electrode and the second electrode is electrically connected to the emitter electrode, and a second one of the first electrode and the second electrode is electrically connected to the first gate electrode.

14. The semiconductor device according to claim 1, wherein a part of the emitter electrode is provided in a contact trench that penetrates the emitter layer from the first main surface side and reaches the base layer, the semiconductor device further comprising a contact layer of the second conductivity type provided at a bottom portion of the contact trench.

15. The semiconductor device according to claim 1, wherein a part of the emitter electrode is provided in a contact trench that penetrates the emitter layer from the first main surface side and reaches the base layer.

16. A method for controlling a semiconductor device, wherein the semiconductor device includes a semiconductor substrate including a first main surface and a second main surface opposite to the first main surface, the semiconductor substrate includes a drift layer of a first conductivity type provided between the first main surface and the second main surface, a carrier accumulation layer of the first conductivity type provided on the first main surface side of the drift layer, a base layer of a second conductivity type provided on the first main surface side of the carrier accumulation layer, an emitter layer of the first conductivity type selectively provided on the first main surface side of the base layer, a buffer layer of the first conductivity type provided on the second main surface side of the drift layer, and a collector layer of the second conductivity type and a cathode layer of the first conductivity type provided on the second main surface side of the buffer layer, the semiconductor device further includes a gate insulating film provided on an inner wall of a trench that penetrates the emitter layer, the base layer, and the carrier accumulation layer from the first main surface side and reaches the drift layer, a first gate electrode provided on the inner wall on the first main surface side with the gate insulating film interposed therebetween, and having an end portion closer to the second main surface than an end portion of the base layer on the first main surface side, a second gate electrode insulated from the first gate electrode, provided on the inner wall on the second main surface side with the gate insulating film interposed therebetween, and having an end portion closer to the first main surface than an end portion of the base layer on the second main surface side, an emitter electrode provided on the first main surface, and a collector electrode provided on the second main surface, the method comprising applying a positive gate voltage to a first one of the first gate electrode and the second gate electrode, determining whether current flows from the collector electrode to the emitter electrode or from the emitter electrode to the collector electrode, while applying the positive gate voltage to the first one, in a case where current flows from the collector electrode to the emitter electrode, applying a positive gate voltage to a second one of the first gate electrode and the second gate electrode, and while applying the positive gate voltage to the first one, in a case where current flows from the emitter electrode to the collector electrode, applying a voltage equal to or less than a reference voltage to the second one.

17. The semiconductor device according to claim 1, wherein the reference voltage is 0 V.

18. The method according to claim 16, wherein the reference voltage is 0 V.
